Problem Statement and Goals
Natural Control

Group 7
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Table 1: Revision history.

Date	Developer(s)	Change
2026-09-23	Aaron	AI-assisted draft.
2026-09-28	Curtis	Reviewed draft for alignment with rubric and expectations.
2026-10-02	Aaron / AI assistance	Draft revisions for peer-review issues 11 and 12; narrow initial scope.

1 Problem Statement

Visually impaired individuals can face many difficulties when navigating screen-based websites and applications that rely on visual feedback. These barriers become especially challenging when completing complex, multi-step digital tasks that require the interpretation of changing system states. Natural Control is a voice-based interface to a computer-use agent aiming to help visually impaired users complete digital tasks while receiving accessible feedback and retaining control.

1.1 Problem

Complex and unfamiliar tasks can require users to navigate multiple applications, interpret changing interface and application states, recover from errors and unexpected outcomes, and determine if actions were completed successfully. Managing these tasks can be particularly difficult without relying on visual feedback.

For instance, arranging an appointment may involve finding availability on a website, checking a calendar, and sending an email. The user must track information across these applications and confirm that the appointment was successfully scheduled.

Barriers like these can make it difficult for users to complete multi-step tasks independently and to remain aware of what's happening while a task is being performed. This problem has become

increasingly important as many activities depend more on websites, applications, and other digital services. Difficulties with nonvisual interactions can therefore limit many users' abilities to interact with the modern digital world.

1.2 Inputs and Outputs

The intended input is the individual's voice as a form of control. Spoken input includes goals, preferences, clarification answers, corrections, approvals, and stop requests. Users who are able may also supply keyboard or text input for assistance.

The system also receives relevant digital-environment context: connected-account data, browser page content and state, and the results of previous tool actions. Available context depends on the connected tools and permissions granted by the user. Browser context may include structured page information and screenshots; it is not limited to information supplied through the user's voice.

The intended outputs are actions on permitted connected accounts and websites, concise spoken progress and results, requests for clarification or approval, and an accessible, reviewable task history. Completion feedback must distinguish verified results, partial work, and unresolved failures.

1.3 Stakeholders

Visually impaired individuals
: Primary stakeholders are visually impaired, English-speaking individuals who want to complete digital tasks while retaining control through non-visual interactions. Initial evaluation will recruit blind Mac users with screen-reader experience; the broader stakeholder group does not imply evaluation of every user population.

Users Wanting Non-Visual Interactions
: Secondary stakeholders are non-visually impaired individuals who still want the ability to complete digital tasks without relying on visual feedback. They may be unable to view their computer screens for other reasons, or may simply have a desire not to rely on their screen.

1.4 Environment

The expected environment is a Mac computer running macOS with Google Chrome, a microphone, audio output, and internet access. Initial supported tasks are limited to browser interaction and connected-account tools. Native Mac application and general local-file control are deferred. Users may also have existing accessibility tools like VoiceOver and keyboard navigation enabled.

2 Goals

The Development Plan, Section 9.3, specifies provisional targets and the evaluation method for these goals. The goals below describe intended outcomes, not achieved results.

G1. **Complete complex digital tasks independently.** Within the initial browser and connected-account scope, users should be able to complete common, multi-step tasks without relying on visual feedback or sighted assistance. Success can be measured by task completion rate and the amount of sighted assistance that is required.

G2. **Maintain awareness and control during digital tasks.** Users should be able to understand relevant task progress and state changes, provide input when required, and stop or redirect tasks when necessary. Measure successful corrections, explicit stop acknowledgment and execution-stop confirmation, and approval before consequential actions. Clarification counts provide context; fewer questions alone do not establish better control.

G3. **Accurately receive results and execute commands.** Users should be able to have confidence that the program is correctly tracking the system's state, retrieving results, and executing their desired actions and commands. Compare reported outcomes and executed actions against independently verified application state and the user's request. Report incorrect or unsupported completion claims separately from successful actions.

G4. **Recover from errors and unexpected outcomes.** Users should be able to identify unexpected, misunderstood, or failed actions and recover without restarting the entire task or relying on sighted assistance. Measure whether recovery succeeds without sighted help or a full restart, along with recovery time and clarification turns.

G5. **Provide a straightforward and reliable setup process.** Individuals responsible for setting up the interface should have a smooth setup process. Routine use after initial setup should not require assistance from the setup helper. Success can be measured by the number of clicks or commands needed, the setup time before the program can be used, as well as how many times the setup helper is required to assist after installation.

3 Stretch Goals

SG1. **Support additional applications and operating systems.** If time permits, extend support to native applications, local-file tasks, additional browsers, and operating systems to make the system accessible to more users.

SG2. **Support a second voice channel.** A useful extension is to allow users to access the same task context through their computer as well as their cell phone. This extension could allow them to interact with their computer through voice-controlled interactions without being directly in front of it.

SG3. **Support saving workflows.** Allowing users to save common workflows, such as paying one's credit card bill, would be helpful and improve their efficiency in interacting with the computer. Users could have the option to complete a task or interaction and save it to a collection of frequent workflows that they can quickly request when desired. Saving a workflow does not grant standing approval: consequential actions still require confirmation of their current details.

4 Custom Documents (CDs)

Two custom documents are proposed to be completed over the course of the project's development.

- **Fall: User/Stakeholder Interview Report.** Record recruitment, participant needs, methods, findings, and resulting design decisions. Separate direct evidence from team hypotheses.

- **Winter: Usability Report.** Evaluate independent completion, awareness, correction, and user control against participants' usual workflows. Report limitations and unsuccessful cases as well as benefits.

Appendix A: Reflection

1. **What went well while writing this deliverable?** Overall, the team communication was well-established and effective. There was consistent communication within the group's Discord server about the deliverable, how the initial draft would be completed, and how review would be handled. The process of using Git for branches and GitHub Pull Requests to foster discussion on changes was also organized and easy to manage.
2. **What pain points arose and how were they resolved?** The group had initially determined that transparently using AI assistance for writing the draft of the document before manually reviewing would be effective and efficient, but Curtis found this to be rather time-consuming when it came down to the review process. Some sections didn't fully align with expectations, were slightly difficult to understand, and were rather vague due to the nature of LLM's and their writing style, so the next process to try moving forward will be primarily manual writing first, then use AI assistance to proofread and expand on anything needing elaboration. This should be beneficial as a whole for the group's effectiveness when it comes to writing proposals and further documents.
3. **How did the team adjust scope?** During the team's planning, the scope was limited to only the visually impaired, as well as only handling web page interactions. After a conversation over email with Dr. Spencer Smith, the scope was suggested to be expanded to general users who might desire the project's functionality, as well as ensuring to handle more applications and complex state changes than just creating a simple screen reader. After meeting with Teaching Assistant Rebecca DiFilippo, this scope was confirmed as sufficient as of the initial time of writing; the team was reminded, however, to be open to further widening the scope or limiting the scope to a small subset of tasks, according to how the development process moves along.

Appendix B: Generative AI Use Disclosure

Tool and use

1. Aaron Loh: OpenAI Codex was used with GPT-6 Astra, primarily at Medium reasoning and occasionally High (settings reported by Aaron). Codex helped extract questions from course requirements, challenge assumptions, research options, and develop ideas through discussion. It also assisted with drafting and revising text throughout this document, organizing references, and LaTeX/PDF formatting. Aaron made the judgment calls and reviewed the work. During integration, Codex also helped reconcile team revisions, clarify measurements, and check the PDFs. No participant research results are claimed. Codex also assisted with the October 2 revisions addressing peer-review issues 11 and 12 and Aaron's decision to narrow the initial scope to browser and account tasks. Review of these revisions by another team member is pending.
2. Curtis Vos: For each section, after performing a manual review and update of the initial draft, OpenAI's GPT-5.6 Luna was used for general feedback and for grammatical assistance. Each

response was read through fully and worked from if deemed applicable, with no generated responses being copied, and no generated feedback being used without full review.

3. Kevin Kim: OpenAI Codex was used for grammar checks and related wording corrections in the Inputs and Outputs section, Goal 5, and stretch goals. Kevin reviewed the suggested changes before they were applied.

The team maintains responsibility for all submitted work.